1 Quick Review

Eigenvalues and eigenvectors are the scariest sounding subject in linear algebra, but compared to vector spaces, are a much easier concept! Everything you need to know about what an eigenvalue is captured in

$$\mathbf{A}x = \lambda x \implies \mathbf{A}x - \lambda x = (\mathbf{A} - \lambda)x = 0 \implies \det(\mathbf{A} - \lambda\mathbf{I}) = 0$$

where $\mathbf{A}$ is the familiar matrix from the $\mathbf{A}x = b$ system.

In summary, eigenvalues are the scalars λ that satisfy the above statement. Eigenvectors are the vectors x that satisfy that statement.

That's about it! The trickiest part is solving the *characteristic equation*, the equation that results from carrying out the determinant. There is no general way to solve such an equation; practice is the only way to get better at solving eigenvalue problems! Make sure you are familiar with the *rational root theorem* and *polynomial division*, they'll be your best friend over the next few weeks.

However, things get a little more interesting when we begin to apply our eigenvalues and eigenvectors. The applications are limitless, but let's take a look at our very first form of decomposition: *diagonalization*!

Diagonalization is a process by which we take a matrix A and split it into the product of three matrices:

$$\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}$$

In practice, we say that the two matrices are *similar*, meaning that they exhibit many of the same properties/behaviors.

The $\mathbf{D}$ matrix is the simplest matrix to form. It is simply the eigenvalues slotted across the diagonal:

$$\mathbf{D} = \begin{bmatrix} \lambda_1 & 0 & 0 & 0 \\ 0 & \lambda_2 & 0 & 0 \\ 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & \lambda_n \end{bmatrix}$$

The $\mathbf{P}$ matrix, though a bit more involved, is simply the eigenvectors augmented as column vectors:

$$\mathbf{D} = [v_1 \mid v_2 \mid \cdots \mid v_n]$$

where v_i is the eigenvector associated with the eigenvalue λ_i.

As a last note, it is easy to think that decomposing the matrix into this form is more complicated. On first glance, this makes sense. But consider the case where we want to find $\mathbf{A}$ to the sixth or seventh power. We could do the matrix multiplication six or seven times: $\mathbf{AAAAAA}$ or $\mathbf{AAAAAAA}$. Or, if we diagonalize it we would find that we could just do $\mathbf{P}\mathbf{D}^6\mathbf{P}^{-1}$ or $\mathbf{P}\mathbf{D}^7\mathbf{P}^{-1}$ gets us the same thing. Think about it, why would $\mathbf{D}^7$ be easier than $\mathbf{A}^7$?

2 Problems

1. Given a set of numbers $S = \{5, 6, 9, 1\}$, which of these are eigenvalues of the following matrix $\mathbf{A}$.

$$\mathbf{A} = \begin{bmatrix} 5 & 0 \\ 6 & 9 \end{bmatrix}$$

$\lambda =$ ____________

2. **(PROOF).** Prove that if a matrix $\mathbf{A}$ has an eigenvalue $\lambda = 0$, then it *cannot* be nonsingular.

3. Let $\mathbf{A}$ be the matrix defined below. **(Adapted from Chapter 5.1 #9).**

$$\mathbf{A} = \begin{bmatrix} 6 & 3 & -8 \\ 0 & -2 & 0 \\ 1 & 0 & -3 \end{bmatrix}$$

   (a) Find the eigenvalues.

(b) Find the eigenbasis for each eigenvalue.

(c) Is this matrix invertible? ____________

4. Recall that functions can be vectors. If functions can be vectors, then we should be able to solve eigenvalue problems with them. In this context, we call these problems *eigenfunction* problems.

   (a) Algebraically, how is an eigenfunction represented? For example, we represent a traditional eigenvalue as any value λ that satisfies $\mathbf{A}x = \lambda x$.

   (b) Let there be an operator A that operates on a function and returns the square of the function. For instance, if $f(x) = x$ then $Af(x) = x^2$. What are the eigenfunctions of this operator?

   (c) Is A a linear operator? ____________

5. Eigenvalues are **EVERYWHERE**. I could never do them justice, but lets's take a look at just one application of them: (All information collected with the help of **White, R. T., & Ray, A. T. (2021).** ***Practical Discrete Mathematics.*** **Packt Publishing.**)

   In data analysis, it is not rare that we analyze data that has an extremely high number of variables. For example, maybe we are trying to calculate the state of 100 air particles, you can imagine that there are many, many variables! However, it is very rare that we actually need that many variables to accurately depict a state (such as the temperature) of the system. Principal Component Analysis (PCA) is one way of extracting new relationships that reduce the overall number of variables.

   After establishing how all the variables in a dataset relate to one another, we can find the *covariance matrix*, $\mathbf{C}$. Let's assume that our dataset produces the following covariance matrix:

$$\mathbf{C} = \begin{bmatrix} 0 & 1 & -2 \\ 1 & 2 & 1 \\ -2 & 1 & 3 \end{bmatrix}$$

(a) Find a matrix $\mathbf{\Lambda}$ where the eigenvalues of $\mathbf{C}$ lie along the diagonal. Sort the eigenvalues into numerical order.

(b) Find a matrix $\mathbf{V}$ that consists of the eigenvectors corresponding to the eigenvalues of $\mathbf{C}$.

(c) Find the matrix product $\mathbf{C}v_1$. This is the first principal component of our new set

(d) For our purposes we only need to explain 50% of the variation in our dataset. This percentage is calculated by applying

$$\frac{\sum_n \lambda_n}{\sum_i \lambda_i} \times 100\%$$

where i is the number of eigenvalues and $n \leq i$. To reach at least 50% what must n be?